

PhysicsAR: Problem Set Reality

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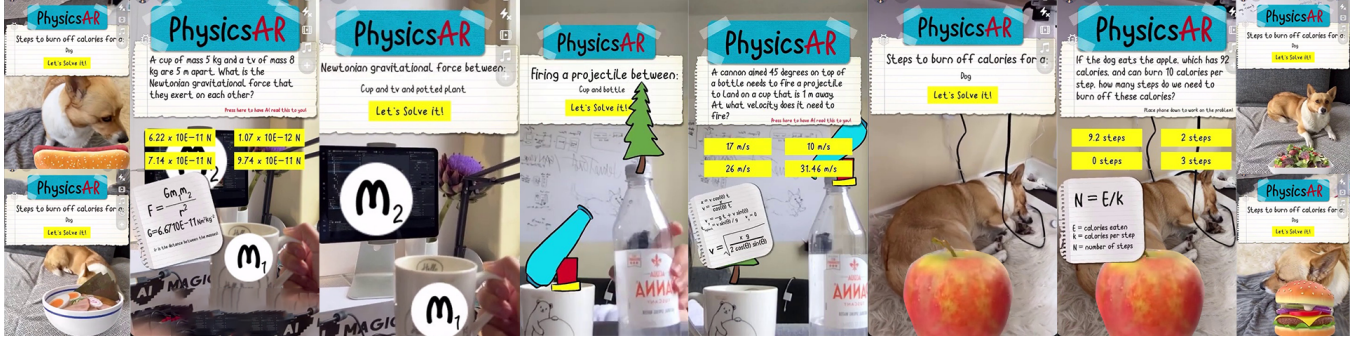


Figure 1: Augmented problem sets from gravitation, projectile motion, and nutrition calculus.

ABSTRACT

PhysicsAR is an augmented reality (AR) application that transcends traditional teaching methods by enabling users to delve into the intricacies of physics through interactions with real-world objects. By seamlessly integrating AR technology with advanced physics concepts, PhysicsAR offers an unparalleled learning experience. This paper presents a comprehensive overview of PhysicsAR, emphasizing its ability to turn real world objects into interactive problems, and the potential it holds for physics education.

CCS CONCEPTS

• **General and reference** → Measurement; **Design**; Evaluation; • **Computer systems organization** → Real-time system specification; Real-time operating systems; Real-time languages; • **Software and its engineering** → Virtual machines; Real-time systems software; Integrated and visual development environments; Object oriented frameworks; Software as a service orchestration system; Genetic programming; Software prototyping; • **Human-centered computing** → **Mobile phones**; **Mobile devices**; **Tablet computers**; **Smartphones**; **Ubiquitous computing**; **Mobile computing**; **Ambient intelligence**; **Ubiquitous and mobile computing systems and tools**; • **Applied computing** → **Computer-assisted instruction**; **Distance learning**; **Interactive learning environments**; **Collaborative learning**; **E-learning**; **Computer-managed instruction**; **Learning management systems**; **Physics**; **Engineering**; **Chemistry**; **Earth and atmospheric sciences**; •

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Information systems → World Wide Web; **Multimedia information systems**; **Spatial-temporal systems**; **Mobile information processing systems**; **Collaborative and social computing systems and tools**.

KEYWORDS

Physics Education, Computer Vision Object Detection Frameworks

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1 INTRODUCTION

Physics education has challenges in teaching physical concepts and having students drill enough problems to understand the problem types and be able to generalize the problem solving skills to apply to problems in the wild. In modern environments, we simply do not have enough time to go through the classic working through textbook problem sets.

[Roger Freedman 2007] [David Halliday 2021] [Herbert Goldstein 2001] [Chang 2015] [Chang 2005] [Chang 2003]

PhysicsAR harnesses the power of AI and AR to create a captivating learning environments by turning everyday objects into instant physics problems. By using computer vision object detection and other algorithms, PhysicsAR seamlessly identifies and tracks real-world objects, facilitating profound explorations of physical phenomena through direct realtime interaction.

Moreover, PhysicsAR embraces the social dimension of learning by fostering collaboration, and knowledge sharing, in the forms of social media-friendly videos and snapshots.

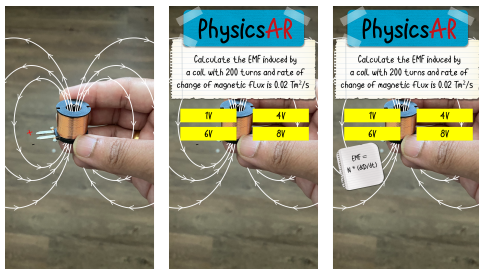


Figure 2: (Left) Magnetic field lines augmented on a solenoid coil. (Center) Problem sets illustrating concepts on magnetic induction. (Right) On-demand hints.

2 THE CORE OF PHYSICS EDUCATION—AND ITS AGE-OLD PROBLEM

Physics education at the K-12 level is fundamentally about teaching a specific technical language and problem solving skills in context of such language. For example, the word “normal” means something very explicit in the language of physics education, but outside of which, could be taken for as all kinds of ambiguity.

In terms of problem solving skills, for example, kinematics equations at the pre-calculus level is primarily about popularization of the classic “classical mechanics” equations of $y = \frac{1}{2}gt^2 + v_{0y}t$ etc (maybe with a few ad hoc geometry-based derivations and empirical data curve matching), whereas at the calculus level, one derives the more elegant series starting from constant gravity: $g = \frac{dv}{dt} \Rightarrow v(t) = \int g dt = gt + v_{0y}$ and $v_y = \frac{dy}{dt} \Rightarrow y = \int v(t) dt = \frac{1}{2}gt^2 + v_{0y}t$

The process of breaking velocity into the Cartesian x and y components itself is rife with ambiguity—and the misunderstanding or lack of practice (or understanding of the basic trigonometry required) leads to early discouragement in learning Physics and cascading loss of self-esteem, to turn Physics class into one of those dreaded requirements one would rather forget.

Using the proper ambient computing software paradigm, we can solve this problem. For instance, students can point their devices at inclined planes, observing the overlay of virtual vectors representing forces like gravity and friction. They can manipulate the angle of inclination or the coefficient of friction, instantly visualizing the impact on the object’s motion and deepening their understanding of mechanical principles.

In more esoteric domains that one would not normally encounter in real life, such as optics, electromagnetic theory, and quantum mechanics, PhysicsAR provides students with superimposed virtual representations of quantum interactions, invisible field lines, and induced effects, enabling them to understand complex concepts in an intuitive and visual way.

2.1 “The Human LLM”: Physical Intuition via Problem Matching Skills

LLMs are perhaps best summarized as generalized pattern matching-solving machines – when exposed to enough body of training, it just gets better. The same general idea applies to students learning physics: the more problems they are exposed to, the more problems they practice, the better they get, and eventually the tricks and

steps learned to solve a particular category of problems generalize to others.

This tailored practice could expand the student’s intuition into further understanding of physics and other areas of science.

3 PROBLEM SET REALITY

The idea of Problem Set Reality is that you can turn anything in reality into infinite problem sets, hence the name!

Some concepts in physics can only be understood from solving problems, and also, physics education is evaluated based on one’s ability to solve problems. Physics problem solving at the high school level is dry and boring and stuck in the textbook.

This is one of the main problems PhysicsAR aims to solve – where we take problem sets into reality, ambient computing, so that you can have your augmented reality app as a physics tutor to practice physics anywhere on things you see in real life.

3.1 Object Identification

Identify objects you see around you, and get a problem set that “works” on the objects. Optionally specify the area of study you’re interested in for problem sets from that sector.

3.2 Robust Problem Set Generation System

The system is by definition set to be able to generate a lot of different problem sets, where the objects are taken as input for problem set generation. Variations to both language and style of problem presentation are added so that the same concept can be presented as many different problems.

4 POST-PANDEMIC EDUCATION INNOVATIONS

4.1 Snap Camera Kit

The current version of PhysicsAR available as a Snap Filter. For the next iteration, we have plans of using Snap’s Camera Kit to create a standalone app that would be made available to download from Apple AppStore / Google PlayStore.

4.2 Unity OpenCV

We can also have this app be a Unity iOS / Android / x-platform app using OpenCVForUnity. This will also allow us to easily have GPT-4 synthesize its re-phrasing of problems.

4.3 Social Media Share-ability of Features

What if learning physics and creating problem sets from your reality can become the coolest thing to share on social media?

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